2.6 Trig Identities Worksheet #2

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Prove each of the following identities

1) $cos\theta \times tan\theta = sin\theta$

2) $\frac{cot\theta}{tan\theta} = \frac{1-sin^2\theta}{1-cos^2\theta}$

3) $cot^2\theta = \frac{cos^2\theta}{1-cos^2\theta}$

4) $\frac{csc\theta}{sec\theta} = cot\theta$

5) $(sin\theta + cos\theta)^2 = 1 + 2\ sin\theta\ cos\theta$

6) $2\ sin^2\theta - 1 = sin^2\theta - cos^2\theta$

7) $\frac{1}{sin^2\theta} + \frac{1}{cos^2\theta} = \frac{1}{sin^2\theta\ cos^2\theta}$

8) $cos^2\theta = sin^2\theta + 2\ cos^2\theta - 1$

9) $tan\theta = tan^2\theta \times cot\theta$

10) $sec^2\theta + csc^2\theta = sec^2\theta \times csc^2\theta$

11) $\frac{1}{1+sin\theta} + \frac{1}{1-sin\theta} = 2\ sec^2\theta$

12) $tan^2\theta - sin^2\theta = sin^2\theta \times tan^2\theta$

13) $\frac{1+2sin\theta\ cos\theta}{sin\theta+cos\theta} = sin\theta + cos\theta$

14) $\frac{sec\theta+1}{sec\theta-1} + \frac{cos\theta+1}{cos\theta-1} = 0$

15) $\frac{csc\theta}{csc\theta-1} + \frac{csc\theta}{csc\theta+1} = 2\ sec^2\theta$